Standard (Stage 2A)

You are one of two firms producing an identical product. You and the other firm simultaneously choose how many units to produce, and neither firm sees the other's choice. The market price depends on the total: P = 109.2 - 1.0 * (your quantity + their quantity). Each unit costs you 10.0 to make, so your profit is (P - 10.0) multiplied by your quantity. Choose your quantity.

equilibrium: produce 33.1

Disguised (Stage 2A)

You and one other operator each draw water from the same reservoir. You both choose how many channels to open at the same time, and neither of you sees the other's choice. The amount each channel delivers depends on the total number open: V = 109.2 - 1.0 * (your channels + their channels). Each channel you open costs you 10.0 credits to maintain, so what you gain is (V - 10.0) multiplied by your number of channels. Choose how many channels to open.

equilibrium: open 33.1 (same numbers, no game words)

Extreme (Stage 2C)

You and another participant each privately choose a number at the same time. Let A = 2 x (your number) + 4 and B = 2 x (their number) + 4. Your score is 99.2 x A, minus A x A, minus A x B. What number do you choose?

equilibrium: x = 14.53, i.e. amount 2x + 4 = 33.1